

§5-4 Indefinite Integrals

Homework : 2,9,17,27,37,39,43,44.

Indefinite Integral

$\int f(x) dx$ = The most general form of the antiderivatives of f

$$= F(x) + C, \quad \text{where } F'(x) = f(x).$$

Example :

$$(i) \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad n \neq -1$$

$$(ii) \int e^x dx = e^x + C.$$

$$(iii) \int \sin x dx = -\cos x + C.$$

$$(iv) \int \frac{1}{x} dx = \ln|x| + C.$$

$$(v) \int \frac{1}{1+x^2} dx = \tan^{-1} x + C.$$

$$(vi) \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C.$$

The Net change Theorem :

The integral of a rate of change
= the net change

$$\int_a^b F'(x) dx = F(b) - F(a).$$

$$\textbf{Example 1 : } \int \frac{2t^2 + t^2\sqrt{t} - 1}{t^2} dt.$$

$$\textbf{Solution : } = \int \left(2 + t^{\frac{1}{2}} - t^{-2} \right) dt = 2t + \frac{2}{3}t^{\frac{3}{2}} + t^{-1} + C.$$

$$\textbf{Example 2 : } \int (10x^4 - \sec^2 x) dx.$$

$$\textbf{Solution : } = 2x^5 - 2 \tan x + C.$$

Example 3 : $\int_{-1}^2 (x - 2|x|) dx.$

Solution : $= \int_{-1}^0 (x + 2x) dx + \int_0^2 (-x) dx$

$$= \frac{3}{2} x^2 \Big|_{-1}^0 - \frac{x^2}{2} \Big|_0^2 = -\frac{3}{2} - 2 = -\frac{7}{2} .$$